XIAOYU (Sarah) ZHANG

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Portfolio: *https://xy19app.github.io/xiaoyu-zhang-portfolio/* | Access code: 1001

Educations

University of Warwick – BSc MORSE (Mathematics, Operational Research, Statistics and Economics) (Sept 2025 – June 2028)

- First Year: **Upper Second Class (2:1)** | Mathematics-focused

The King's School, Canterbury, UK (Sept 2022 – June 2025)

- **A Levels**: Mathematics (A* | 91%), **Further Mathematics (A* | 91%)**, Computer Science (B), Chinese (A*)
- **GCSEs** (relevant to current study): Mathematics (9), Further Mathematics (9), Economics (7), Computer Science (7)

Finance & Professional Experience

CEO Office Intern – Corporate & Institutional Banking (CIB), BNP Paribas (July 2026 – Sept 2026)

- Produced daily report on **market and BNPP client news** for senior managers; **created an automated workflow** using **Hermes**, **reducing production time from 1.5 hours to 20 minutes**, with the prompt handed over to future interns
- Analysed China's AI and healthcare industries, using **Hermes** to efficiently **collect data on around 380 companies**, covering market cap, FY25 revenue, net profit, CAPEX and sanctions flags; then used the dataset to create a **19-page presentation** linking the CEO's AI perspective to capital markets, trade finance and foreign-exchange opportunities
- Cleaned, checked, and verified **85,000+ rows** of cross-border banking data with the PRC coverage team, using **PivotTables**, **VLOOKUP** and **filters**; then used the final dataset to **draw 15 graphs** across countries, industries, and products for a **10-page presentation**
- **Wrote a report** on **US – China sanction lists** and their **impact** on deposits, RMB/USD payment transactions, trade finance, loans, and securities; then **presented the findings directly to the CEO**, who subsequently shared the report with other departments for further actions

Morgan Stanley Step In, Step Up – Women's Insight to Banking (October 2023)

- Explored global capital markets, risk modelling, and investment banking through trading-floor visits and professional workshops
- **Led a team project** to pitch for a simulated **£100,000 angel investment** by developing an online education platform; I allocated tasks, did market research, and contributed to the final investor presentation; networked with industry professionals

Morgan Stanley Early Careers Amplify Trading Event (5 December 2023)

- **Applied macroeconomic knowledge** to stock price movements through simulated trading exercises
- Achieved **4th place** in a competitive trading round by making quick trading decisions under time pressure

School Young Enterprise Competition (Sept 2023 – June 2024)

- Worked in a 15-member team to develop, market and sell **two products**: a school-branded insulated cup and a cocktail recipe book
- Contributed to product research, content writing, and book design, gaining experience in business and teamwork

Economics Society (Sept 2023 – June 2025)

- Discussed global economic news and international trading issues using *The Economist*, *Financial Times*, and *BBC*
- Debated solutions to macroeconomic issues, developing analytical and communication skills

Independent Research & Investment Projects

US AI Value Chain Investment Research (August 2026) *See full research website in my portfolio*

A **research dashboard** helping **individual investors** analyse US AI companies and identify investment opportunities and risks

- **Built** a **database** of **67 US AI companies** using ~110 public sources; grouped them across **9 supply-chain layers** and **3 strategic tiers**
- **Developed investment profiles** for the **top 20 companies**; analysed their growth, competitive advantages, risks, and investment potential
- Analysed how **value is created across the AI supply chain**; found the areas with the **most value and pricing power**

Self-Directed Investment & Trading

- Managed a **personal equity portfolio** on IBKR (Apr 2026 – present), achieving a **+30.7% return** on initial capital (until 4th Sept 2026)
- **Long-term: Built a systematic investment strategy** for the **S&P 500 and Nasdaq 100** based on their historical performance and risk analysis, using predefined investment rules and market adjustments to **reduce emotional decision-making**
- **Short-term:** Made short-term stock trades using a **self-developed target-price framework**, to identify opportunities from market movements and company-specific developments, while applying risk management and disciplined decision-making

John Locke Essay Competition (2024) – Commendation Prize (Top 16.9% globally)

- Wrote "*What is the optimal global population?*", **applying economic theory** and **quantitative modelling**
- **Developed a model** linking habitable land area and population density to estimate the optimal global population
- Collected and analysed per capita GDP and happiness-index data, and applied the Malthusian Growth Model

Skills & Interests

Programming Skills *Explore the app in my portfolio*

- Proficient in **Python**, **Java**, **SwiftUI**, and **HTML**; Currently learning **SQL** and **R**
- Built **4 iOS apps** (one **published on the App Store**), a note-taking webpage, personal memoir website, and a classic Snake game

AI Agent Tools & Automation

- Used **Hermes** and **Claude Code** for market research, data collection, Word and PowerPoint automation, and web development
- **Built my own AI Agent skills**, including a bilingual news-summary skill that **reduced** daily production time from **1.5 hours to 20 minutes**

Mathematical Competitions

- UKMT SMC 2023 & 2024 – Gold award twice (UK top 10%); MOG 2023 – Distinction (Top scoring); BMO 2023 (participated)

Other Skills & Interests

- **Languages**: English (Advanced Proficiency), Mandarin (Native)
- **Technical Skills**: Word, Excel, PowerPoint; LaTeX (4 years – self-taught); Adobe Lightroom and Photoshop; **Blender (50+ projects)**
- **Music**: **16 years of piano** (Grade 10 Shanghai / equivalent to **UK Grade 9**), 4 years of violin, 1 year of guitar; **20+ original compositions**
- **Sports**: 2 years of horse riding, 1 year of fencing